

Recap on Newton's 3 Laws of Motion

SS ILP Physics (Intermediate), Week 3 Class 1



Outline

- Recap on Newton's three Laws of Motion
- Practice Questions

Newton's 1st Law of Motion

- An object remains its state of rest or uniform motion at a constant velocity unless it is acted upon a non-zero resultant force.

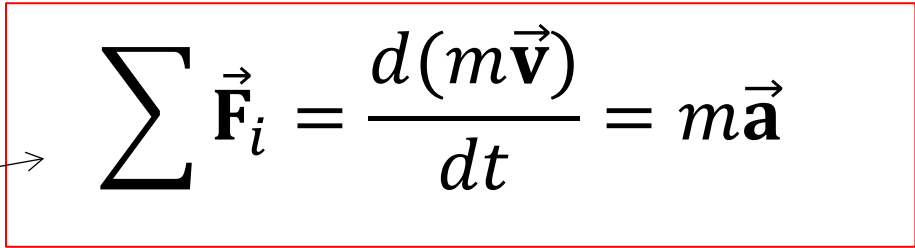
$$\sum \vec{\mathbf{F}}_i = \vec{\mathbf{0}} \rightarrow \vec{\mathbf{v}} = \text{constant}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law>

Newton's Second Law

- The acceleration is proportional to the resultant force, in the same direction and inversely proportional to the mass,

*Sum of all the forces
acting on that object


$$\sum \vec{F}_i = \frac{d(m\vec{v})}{dt} = m\vec{a}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-3/Newton-s-Second-Law>

Newton's Third Law

- To every action, there is always an opposite reaction of the same magnitude,

$$\vec{F}_{2,1} = -\vec{F}_{1,2}$$

force on object 2 due to object 1

force on object 1 due to object 2

<https://www.physicsclassroom.com/class/newtlaws/Lesson-4/Newton-s-Third-Law>

Solving dynamics problems with Newton's three laws

- I. Identify the object(s) of interest.
- II. Draw free body diagram(s) of the object(s).
- III. Choose a proper coordinate system and determine the components of the forces and **acceleration** along coordinate axes.
- IV. Apply Newton's 2nd law of motion to write equations with unknowns,

$$\sum_i^N \vec{F}_i = m\vec{a}$$

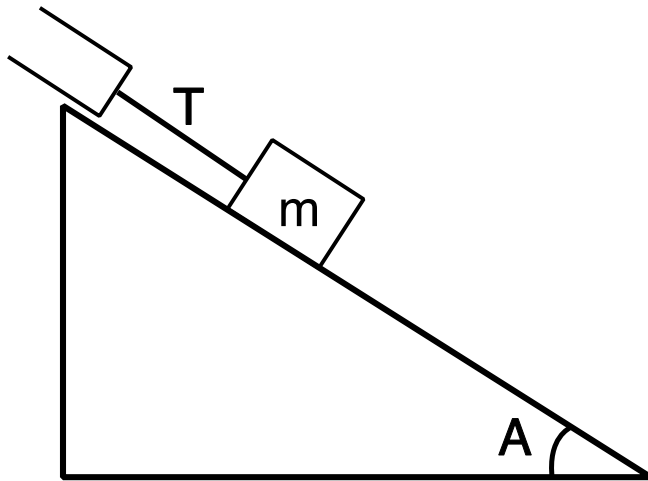
- V. In addition to Newton's 2nd law, identify any other equations you might need for solving unknowns.

H. D. Young and R. A. Freedman, University Physics (14th edition), p160.

Case problem 1

Consider the following system:

- a) If the mass is $5kg$ and $A = 50^\circ$, what is the tension, T ? Assume no frictional forces and the block is motionless.
- b) If the force of tension is $25N$, the force of static friction is $40N$, the block has a mass of $10kg$, and the block is motionless, what is the angle A ? ([link](#))



$$g = 10 \frac{m}{s^2}$$

Case problem 1: Solution

- a) Since there is no friction between the mass and slope, there are only two relevant forces acting on the mass: gravity and tension. Furthermore, since the block is not in motion, we know

$$F_{net,x} = 0 \qquad T - mg \sin A = 0$$

Substituting in an expression for the force of gravity, we get:

$$T = mg \sin(A) = 5kg \times \frac{10m}{s^2} \times \sin(50^\circ) = 38.3N$$

Case problem 1: Solution

b) There are three relevant forces acting on the block in this scenario: friction, tension, and gravity. We are given two of these values, so we simply need to develop an expression for the force of gravity in the direction of the slope. Since the block is motionless, we can write:

$$F_{net,x} = 0 \qquad mg \sin A - T - f_s = 0$$

Substituting in an expression for the force of gravity, we get:

$$mg \sin(A) = T + f_s$$

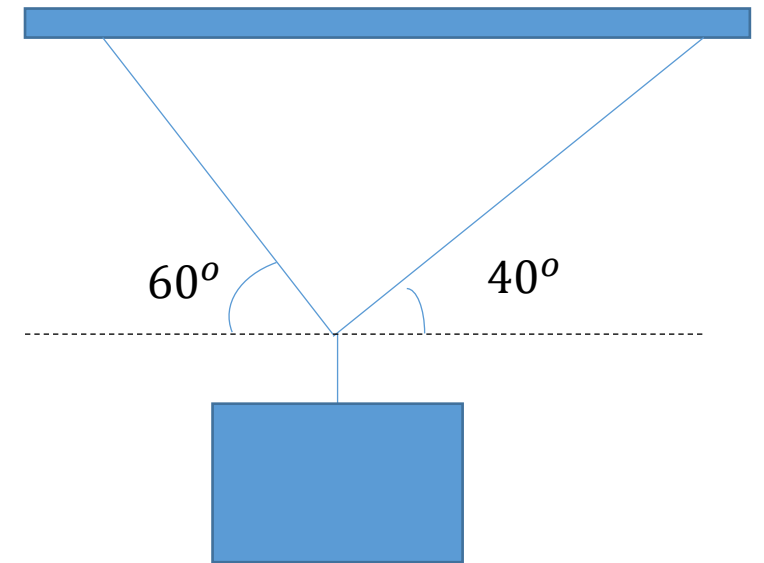
Rearrange to solve for the angle:

$$\theta = \sin^{-1} \left(\frac{T + f_s}{mg} \right) = \sin^{-1}(0.65) = 40.5^\circ$$

Case problem 2

Two ropes are connected to a steel cable that supports a hanging weight as shown in the figure.

- a) Draw a free-body diagram showing all the forces acting at the knot that connects the two ropes to the steel cable. Based on your force diagram, which of the two ropes will have the greater tension?
- b) If the maximum tension either rope can sustain without breaking is 5000 N, determine the maximum value of the hanging weight that these ropes can safely support. You can ignore the weight of the ropes and the steel cable.



H. D. Young and R. A. Freedman, University Physics (14th edition): 5.61, p189

Case problem 2: Solution

We can apply Newton's 2nd law to the knot as it is not accelerating: $\sum_{i=1}^N \vec{F}_i = m\vec{a}$

From the free body diagram sketched we can derive a set of equations relating the weight W to the tensions. T_1 is more vertical, so supports more of the weight and is larger. You can also see this from:

$$\sum F_x = ma_x \quad \Rightarrow T_2 \cos(40) - T_1 \cos(60) = 0 \Rightarrow T_2 = \frac{T_1}{\cos(40)/\cos(60)}$$

T_1 is larger so set $T_1 = 5000N$. Then $T_2 = \frac{T_1}{1.532} = 3263.5N$

$$\sum F_y = ma_y \quad \Rightarrow T_1 \sin(60) + T_2 \sin(40) = T = w = 6428 N$$

Case problem 3

A car is driving on a level surface with a coefficient of static friction of $\mu_s = \frac{3}{4}$ and a coefficient of kinetic friction of $\mu_k = \frac{2}{3}$.

Determine the car's minimum braking distance with normal brakes and antilock brakes as a function of initial speed v_0 . How do these two methods of stopping a car compare? ([link](#))

Hints:

- Stopping a car with ordinary brakes may result in wheel lock. When this happens, the tires skid and the car experiences kinetic friction.
- Cars equipped with an antilock brake have a sensor releases the brake pads and re-engages them. If they lock up again, the antilock brake releases the brake pads again. This processes can repeat many times a second. The car is then stopped using mostly the force of static friction.

Case problem 3: Solution

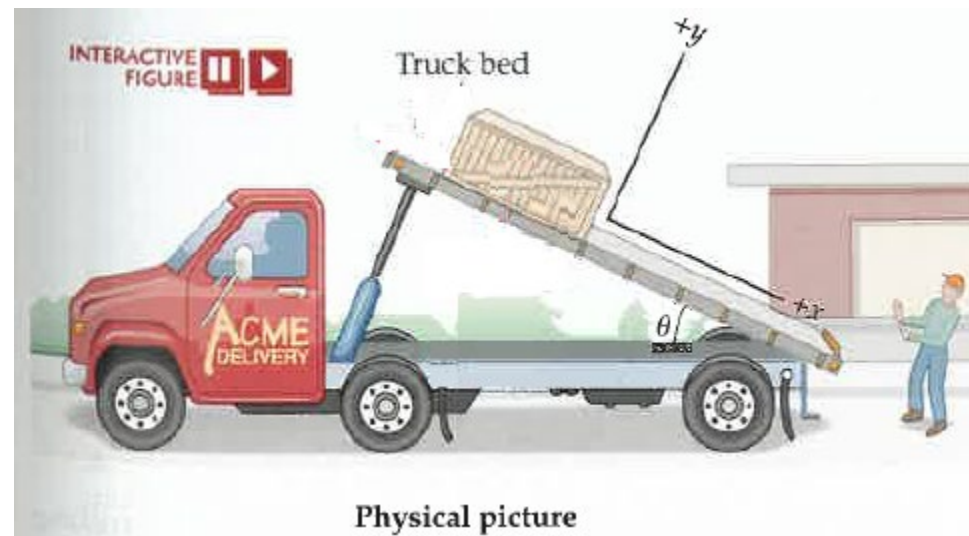
To solve this problem, determine acceleration using the displacement-velocity formula of kinematics. Set this equation equal to the formula for acceleration due to friction derived above.

$$v_0^2 = 2a\Delta s = 2\mu g\Delta s \rightarrow \Delta s = \frac{v_0^2}{2\mu g}$$
$$\frac{\Delta s_{antilock}}{\Delta s_{normal}} = \frac{\frac{v_0^2}{2\mu_s g}}{\frac{v_0^2}{2\mu_k g}} = 0.889$$

Antilock brakes need 90% of the distance of locking brakes to stop a car traveling at the same speed. In addition to reduced braking distance, antilock braking systems also increase performance during extreme braking. Locked brakes are useless for steering. Antilock brakes ensures that the wheels retain their static frictional grip on the road, which allows for manoeuvring while braking in an emergency. ([link](#))

Case problem 4

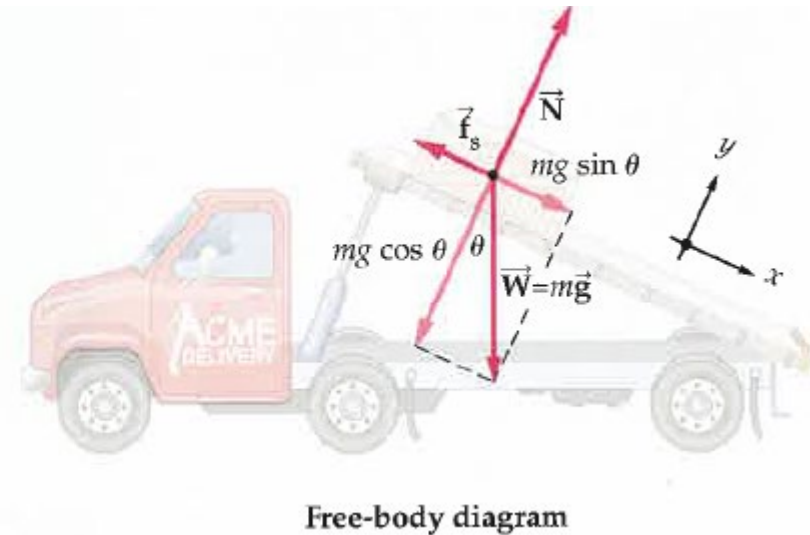
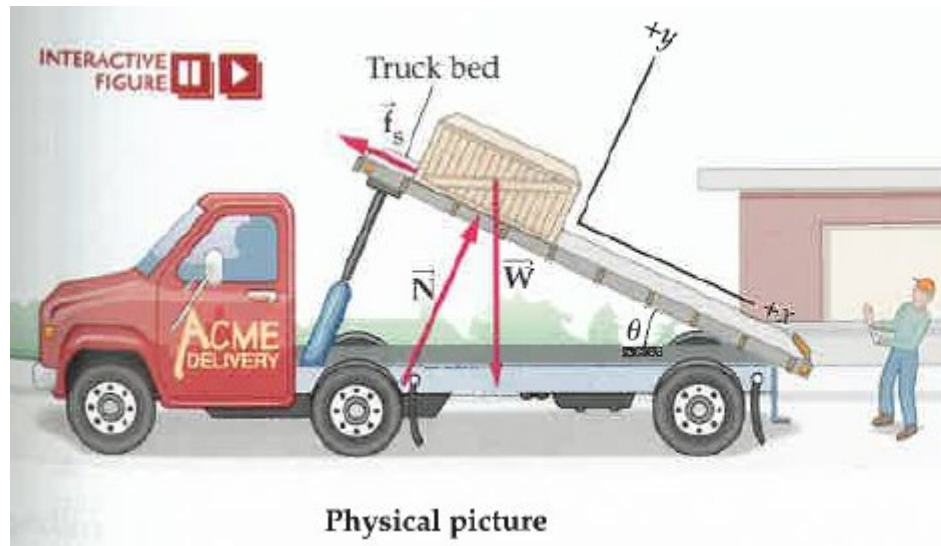
A flatbed truck slowly tilts its bed upward to dispose of a 95.0-kg crate. For small angles of tilt the crate stays put, but when the tilt angle exceeds 23.2° , the crate begins to slide. What is the coefficient of static friction between the bed of the truck and the crate? Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-3, p147

Case problem 4: Solution

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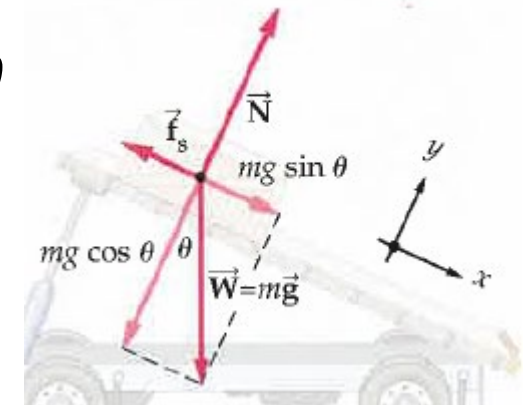
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$$\sum F_x = ma_x \rightarrow -f_s + W_x = 0 \rightarrow -\mu_s N + mg \sin \theta = 0 \rightarrow \mu_s = \frac{mg \sin \theta}{N} = 0.429$$

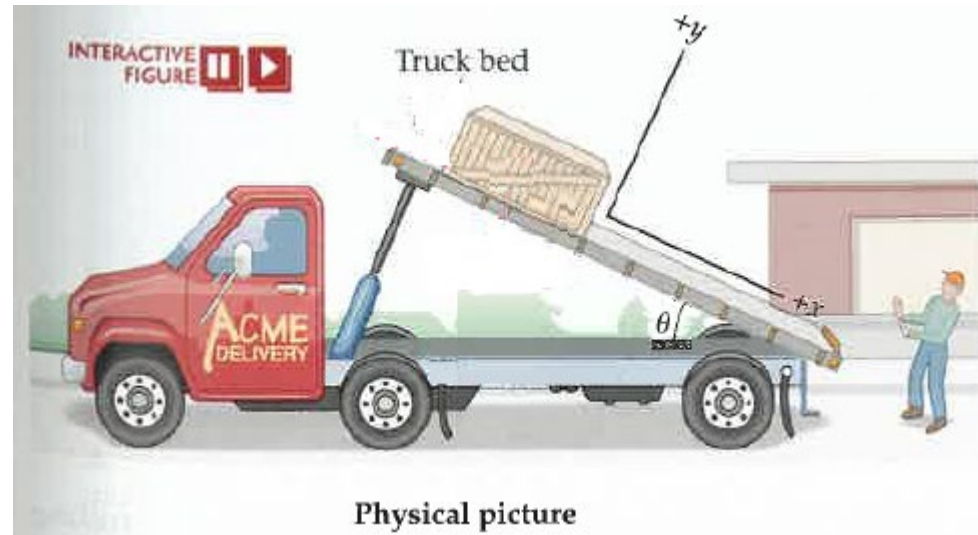
$$\sum F_y = ma_y \rightarrow N - W_y = 0 \rightarrow N - mg \cos \theta = 0 \rightarrow N = mg \cos \theta$$



J. S. Walker, Physics, Example 6-3, p147

Case problem 5

In case problem 4, what is the magnitude of the force of static friction acting on the crate when the truck bed is tilted at an angle of 20.0° . Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Active Example 6-1, p148

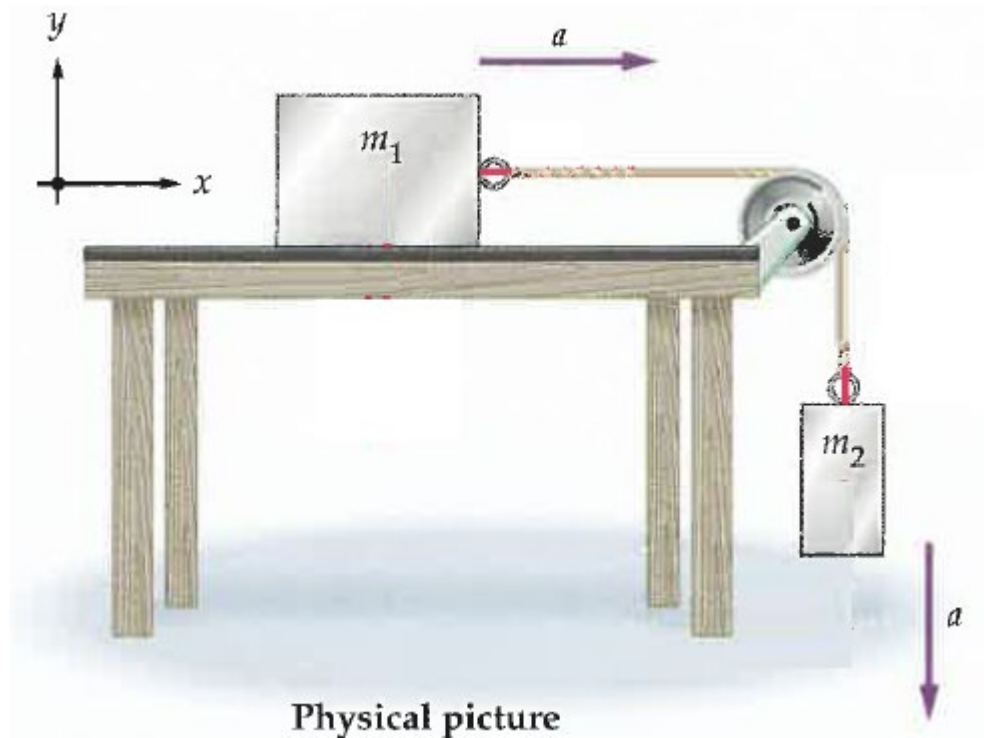
Case problem 5: Solution

In case problem 4, what is the magnitude of the force of static friction acting on the crate when the truck bed is titled at an angle of 20.0° . Take $g = 9.81 \text{ m/s}^2$.

$$\sum F_x = ma_x \rightarrow -f_s + W_x = 0 \rightarrow -f_s + mg \sin \theta = 0 \rightarrow f_s = mg \sin \theta = 319 \text{ N}$$

Case problem 6

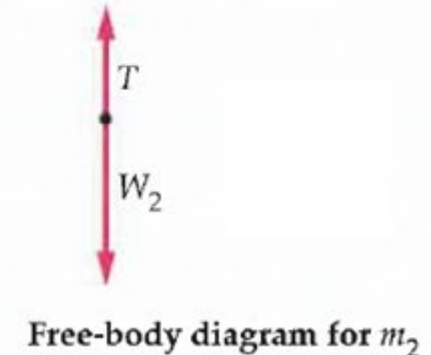
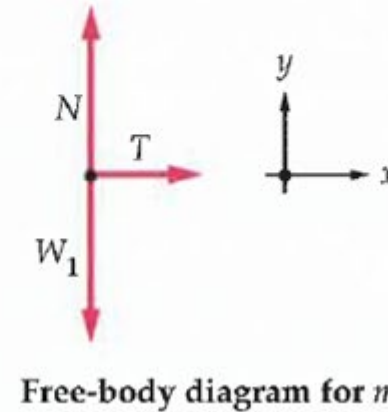
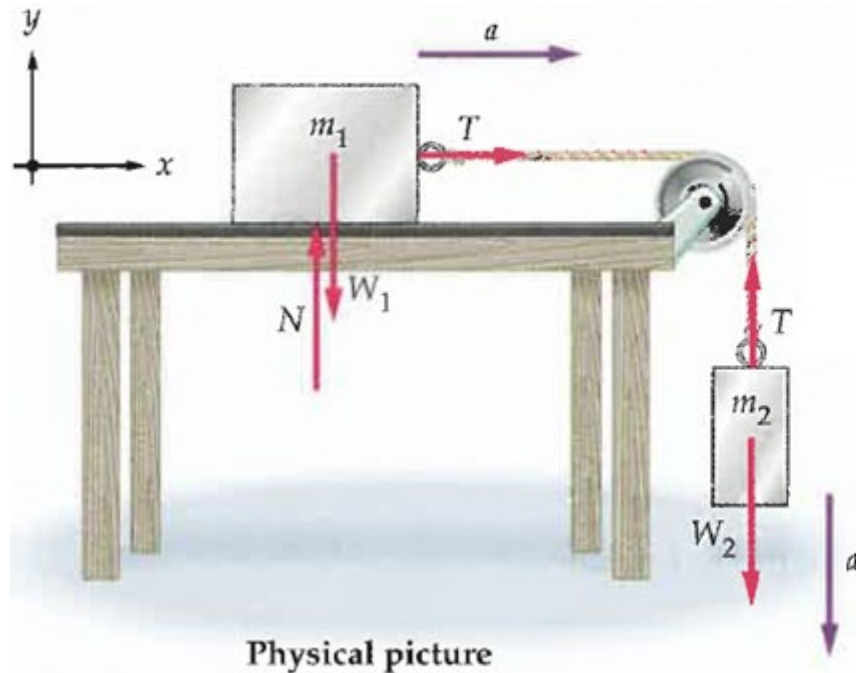
A block of mass m_1 slides on a frictionless tabletop. It is connected to a string that passes over a pulley and suspends a mass m_2 . Find the acceleration of the masses and the tension in the string. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-6, p159

Case problem 6: Solution

A block of mass m_1 slides on a frictionless tabletop. It is connected to a string that passes over a pulley and suspends a mass m_2 . Find the acceleration of the masses and the tension in the string. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-6, p159

Case problem 6: Solution

A block of mass m_1 slides on a frictionless tabletop. It is connected to a string that passes over a pulley and suspends a mass m_2 . Find the acceleration of the masses and the tension in the string. Take $g = 9.81 \text{ m/s}^2$.

$$\sum F_x = ma_x \rightarrow T = m_1 a_1$$

$$\sum F_y = ma_y \rightarrow T - m_2 g = -m_2 a_2$$

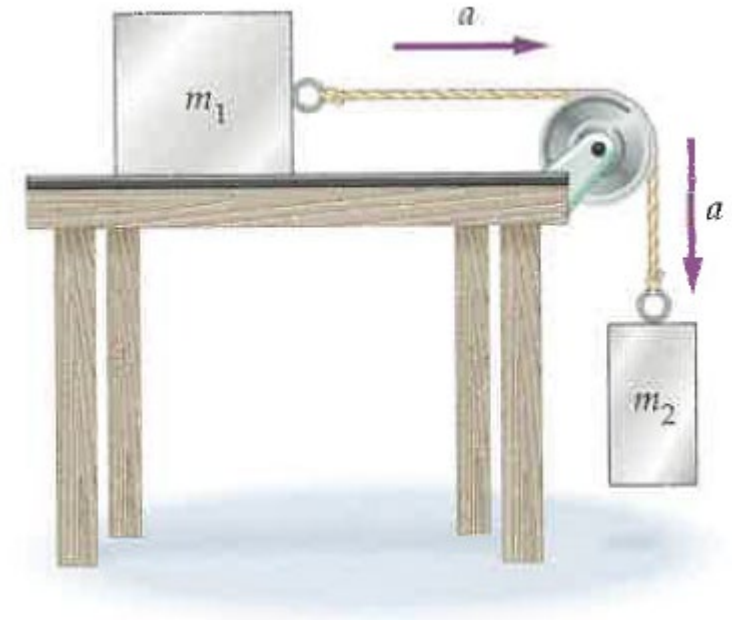
$$a_1 = a_2 = a$$

$$a = \left(\frac{m_2}{m_1 + m_2} \right) g$$

$$T = m_1 a_1 = \left(\frac{m_1 m_2}{m_1 + m_2} \right) g$$

Concept question

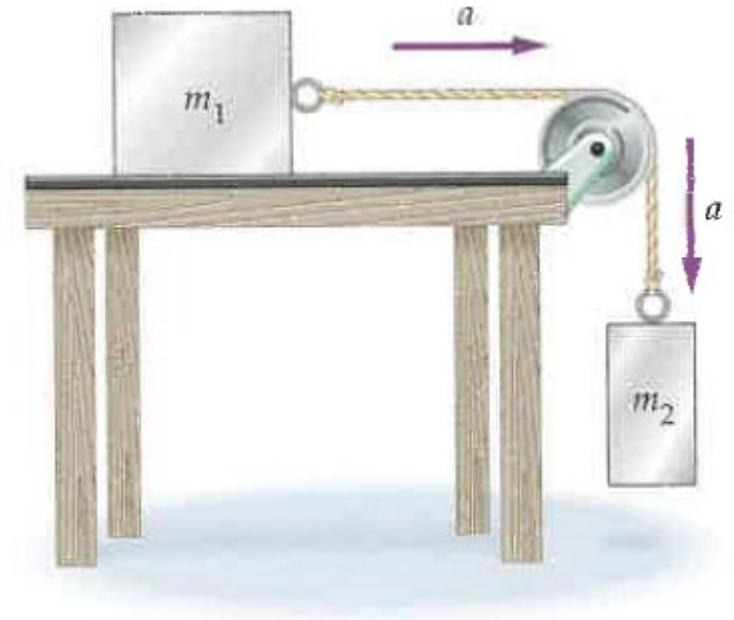
Two masses, m_1 and m_2 , are connected by a string that passes over a pulley. Mass m_1 slides without friction on a horizontal tabletop, and mass m_2 falls vertically downward. Both masses move with a contact acceleration of magnitude a . Is the tension in the string (a) greater than , (b) equal to, or (c) less than m_2g ?



J. S. Walker, Physics, Conceptual Checkpoint 6-4, p159

Concept question: Solution

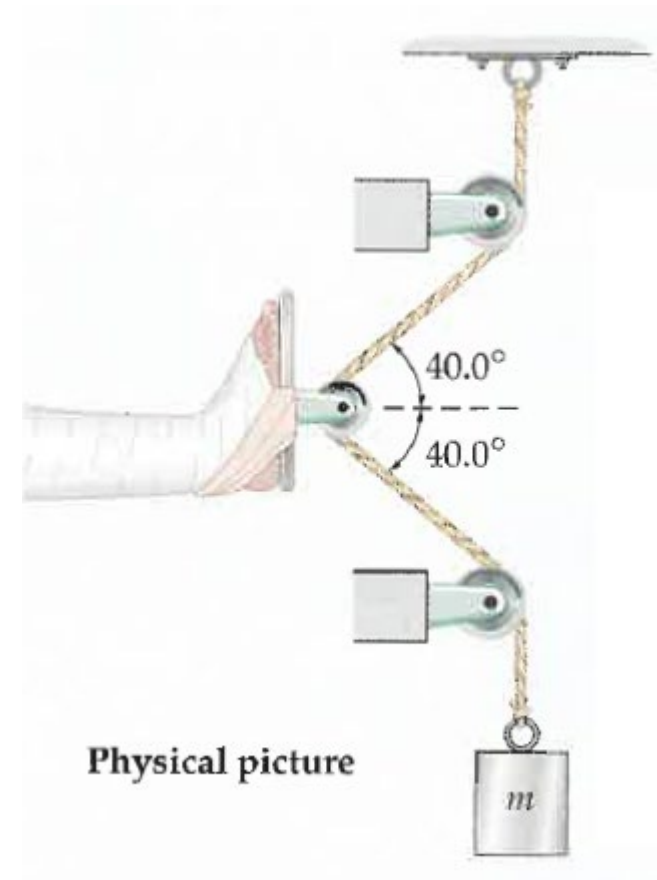
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Off-class practice question(s)

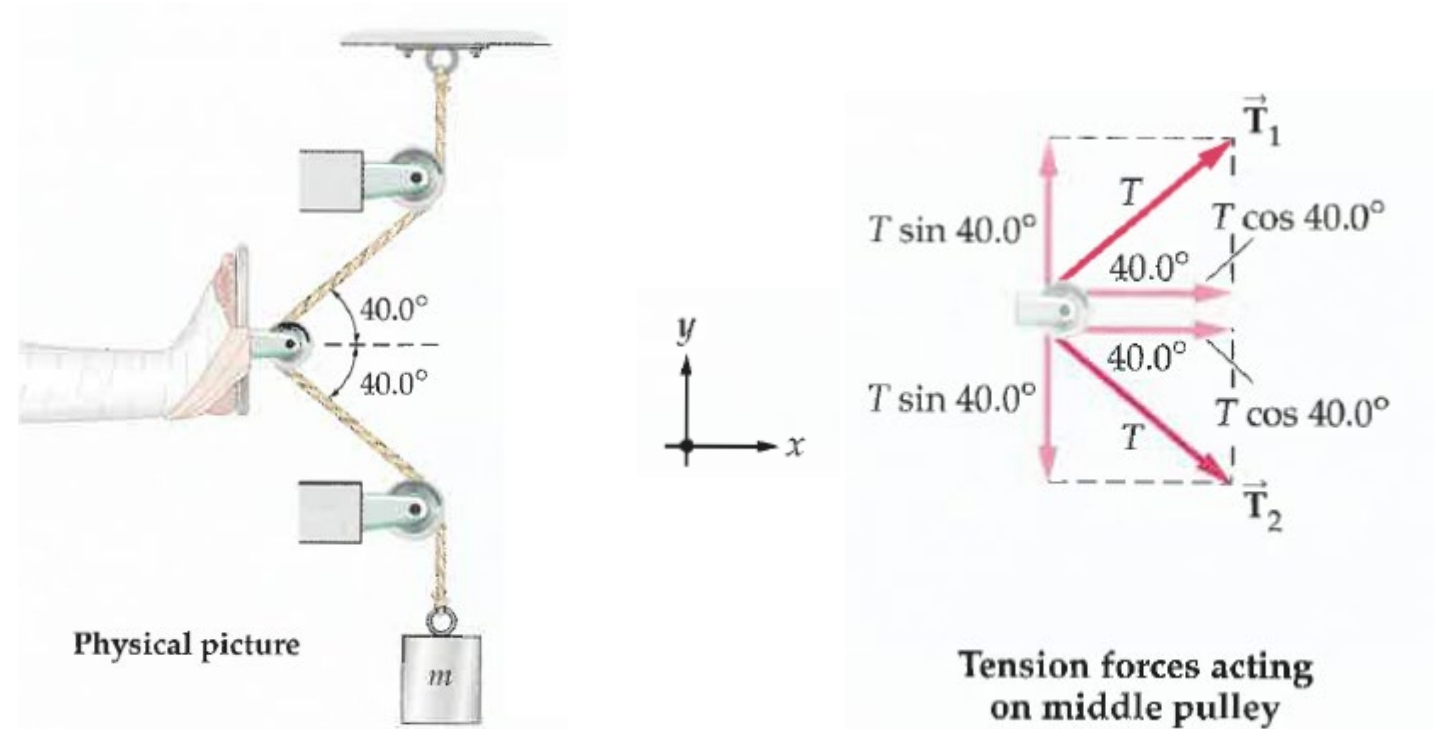
Case problem

A traction device employing three pulleys is applied to a broken leg. The middle pulley is attached to the sole of the foot, and a mass m supplies the tension in the ropes. Find the value of the mass m if the force exerted on the sole of the foot by the middle pulley is to be 165 N. Take $g = 9.81 \text{ m/s}^2$.



Case problem: Solution

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J. S. Walker, Physics, Example 6-4, p151

Case problem: Solution

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$$\sum F_y = ma_y \rightarrow T_1 \sin 40.0^\circ - T_2 \sin 40.0^\circ = 0 \rightarrow T_1 = T_2 = T = mg$$

$$\sum F_x = ma_x \rightarrow T_1 \cos 40.0^\circ + T_2 \cos 40.0^\circ = 165 \rightarrow 2T \cos 40.0^\circ = 165$$

$$2mg \cos 40.0^\circ = 165 \rightarrow m = 11.0 \text{ kg}$$

